

Batch: A-4 **Roll No.:**16010122151

Experiment / Assignment / Tutorial No: 3

TITLE: Introduction to UI/UX design principles.

Objective:

Understand and apply core UI/UX design principles to analyze real-world products and propose user-centered solutions that improve interaction.

Expected OUTCOME of Experiment: CO 2

To design a more seamless and less frustrating interaction experience for everyone with the technologies they encounter daily.

Books/ Journals/ Websites referred:

1. Don Norman, “The Design of Everyday Things” (Book)
 2. <https://www.nngroup.com/articles/ten-usability-heuristics/>
 3. “Interaction Design: Beyond Human-Computer Interaction” by Helen Sharp, Yvonne Rogers, and Jenny Preece (Book)
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Background Theory:

User Interface (UI) and User Experience (UX) design are critical disciplines in computer engineering, focusing on creating products that are not only functional but also intuitive, efficient, and enjoyable to use. While UI pertains to the visual elements and interactive components of a product, UX encompasses the entire journey a user takes when interacting with it. At the heart of good UI/UX design are fundamental principles that guide designers and developers in making user-centric decisions.

Key UI/UX Design Principles:

1. **Visibility:** The more visible functions are, the more likely users will know what to do next. When controls are hidden, discoverability is reduced. Good design ensures the state of the system is always clear to the user.
 - *Example:* Clearly labeled buttons, status indicators (e.g., loading spinners, battery icons).
2. **Feedback:** Every action should have an immediate and appropriate response from the system.

Feedback confirms that an action has been registered and informs the user about the result.

- *Example:* A button changing color on click, an error message appearing after invalid input, a sound confirming an action.

3. **Affordance:** The perceived and actual properties of an object that suggest how it can be used.

A well-designed object "affords" its use.

- *Example:* A door handle affords pulling, a button affords pushing. In digital interfaces, a hyperlink affords clicking.

Signifiers: Any mark or sound, any perceivable indicator that communicates appropriate behavior to a person. While affordances are about what *can* be done, signifiers are about *how* to do it.

- *Example:* An arrow pointing to a scrollable area, a "Click Here" text, a blinking cursor in an input field.

4. **Mapping:** The relationship between controls and their effects. Good mapping leads to intuitive operation, where the spatial or conceptual arrangement of controls corresponds naturally to the effects they produce.

- *Example:* A car's steering wheel mapping directly to the direction of movement; volume controls on an audio device increasing volume when moved up.

5. **Consistency:** Similar operations should have similar sequences of actions, and similar elements should look and behave similarly. Consistency reduces cognitive load and improves learnability.

- *Example:* Navigation menus appearing in the same place across different pages of a website; standard icons having consistent meanings.

6. **Error Prevention and Recovery:** Designs should aim to prevent errors from occurring in the first place. When errors do occur, the system should provide clear, understandable messages and easy ways for users to recover from them.

- *Example:* Disabling a "Submit" button until all required fields are filled; providing an "Undo" option.

7. **Flexibility and Efficiency of Use:** Allow users to tailor frequent actions (e.g., keyboard shortcuts for experts) while also providing clear paths for novices.

- *Example:* Customizable dashboards, keyboard shortcuts, drag-and-drop functionality.

Understanding these principles is crucial for computer engineers, as it enables them to design and build software that is not just technically sound but also user-friendly and effective in meeting user needs.

Laboratory Work:

Activity:

1. Choose two to three different products—some with good UI/UX and others with poor UI/UX—for analysis.

Examples: mobile banking app, social media platform, video streaming service, university portal, or calculator app.

2. Highlight which UI/UX principles are consistently well-implemented across products.
3. Identify which principles are frequently violated or ignored.
4. Suggestions for design improvements and re-design/re-sketch the UI to overcome the flaws or optimize the existing design for better usability (<https://www.nngroup.com/articles/ten-usability-heuristics/>).

1. Chosen Products

- **Burger King App** – Poor UI/UX (too jazzy, cluttered).
- **District App** – Very good UI/UX (modern, clean, intuitive).
- **Somaiya App** – Good UI/UX (simple, functional for students).



2. Well-implemented UI/UX Principles

Across **District** and **Somaiya apps**, some principles are well-followed:

- **Clarity & Readability** – Text and icons are easy to read with proper spacing.
- **Navigation Consistency** – Both apps use bottom navigation bars, making features easy to find.
- **User-centered Design** – District focuses on personalized offers/events, and Somaiya focuses on student needs like attendance and notices.

Example: In the Somaiya app, students can quickly mark attendance or access lecture schedules from the home screen, which reduces effort and improves usability.

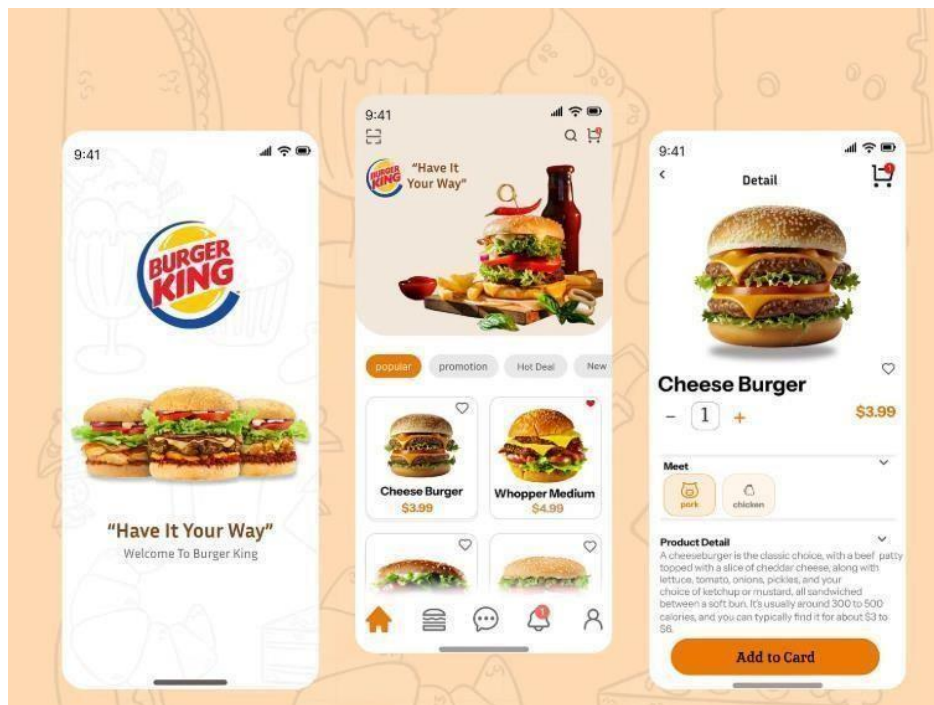
3. Frequently Violated Principles

The **Burger King app** violates several UI/UX principles:

- **Visual Hierarchy** – Too many bright banners and colors compete for attention, overwhelming the user.
- **Simplicity** – Overloaded with promotions and menu items on a single screen, which causes cognitive overload.
- **Consistency** – Fonts, colors, and layouts change too much, making the design feel cluttered.

Example: The homepage immediately throws multiple burger promotions, deals, and offers at the user, instead of guiding them to explore the menu smoothly.

4) Design Improvements & Redesign Suggestions



Applying **Nielsen's 10 Usability Heuristics** [Nielsen Norman Group](#) to the current interface issues and proposed redesign:

1. Aesthetic & Minimalist Design

- **Problem:** Overly jazzy screens with multiple banners and promotions create visual noise.
- **Improvement:** Strip away clutter; only display high-priority items prominently. Use whitespace and grouping to let key actions stand out—like —Order Now|| or —Menu.||
- **Benefit:** Reduces cognitive load and allows users to focus on core actions immediately.
[Nielsen Norman Group+1](#)

2. Visibility of System Status

- **Problem:** Lack of feedback on interactions (e.g., clicks don't show clear responses or loading state).
- **Improvement:** Implement loading indicators (spinners or skeleton screens), confirmation popups, or subtle animations when an action is registered.
- **Benefit:** Signals responsiveness and keeps users informed

3. Recognition Rather Than Recall

- **Problem:** Users struggle to recall where to tap due to inconsistent icons or menus.
- **Improvement:** Use universally recognizable icons (cart, search, menu) and group options logically under clear headings.
- **Benefit:** Reduces memory effort and makes tasks more intuitive.

4. Consistency & Standards

- **Problem:** Font styles, button designs, and layouts change across different screens.
- **Improvement:** Adopt a consistent design system—uniform button shapes, colors, iconography, and layout structures.
- **Benefit:** Enhances familiarity and trust, reducing learning time.

5. Error Prevention

- **Problem:** Users may accidentally tap wrong deals or links due to close spacing and busy visuals.
- **Improvement:** Increase touch target spacing, include confirmation prompts for actions like "Add to Cart," and highlight defaults.
- **Benefit:** Prevents unintended actions and lowers user frustration.

6. Match Between System and the Real World

- **Problem:** Industry jargon (—Crazy App Deals) can confuse users.
- **Improvement:** Use clear, user-friendly labels like —Special Offer or —Combo Deals.
- **Benefit:** Better aligns with users' expectations and reduces confusion.

7. User Control & Freedom

- **Problem:** No clear —back or —undo in promotional sections or deep navigation paths.
- **Improvement:** Emphasize back navigation, close buttons, or breadcrumbs to let users exit easily.
- **Benefit:** Empowers users to explore without fear, improving overall satisfaction.

Design Sketch Concept (Verbal)

- **Header:** Clean, minimal—only logo, location selector, and cart icon.
- **Hero Section:** A single rotating banner for featured deals (less visual noise).
- **Primary Actions:** Clearly grouped buttons—**Order Now, View Menu, Offers.**
- **Menu Categories:** Simple icons with text (e.g., Burgers, Combos, Desserts), evenly spaced.
- **Footer:** Persistent navigation (Home, Offers, Cart, Profile) with consistent icons.

Post Laboratory Questions:

Following questions are to be handwritten and attach photograph of the answer in this document.

1. Analyze how the principles of mapping, affordance, and signifiers are applied in the design of a TV remote or AC remote interface.

→ Mapping : Describes how the arrangement of controls on a device corresponds logically and intuitively to their effects.

o TV Remote :

- The directional arrows (up, down, left, right) are often placed in a circular arrangement, mimicking how users expect to navigate menu or channels.
- Volume and channel buttons typically have "+" and "-" signs and are grouped separately. Users intuitively know pressing "+" increases volume or advances the channel, while "-" does

The opposite -

◦ AC Remote

◦ temperature controls use arrow buttons (usually up/down), mapping directly to temperature increase or decrease.

◦ Mode settings (fan, cool, dry, heat) are arranged in a way that each button explicitly maps to its function.

→ Their spatial & conceptual mapping reduces confusion, as users' mental models of device operation are supported by button placement & labeling.

→ Affordance: The property of a control that suggests how it should be used.

◦ TV Remote :-

◦ Physical buttons afford pressing. Their shape, tactile feedback, and presence indicate to users that they're meant to be pushed.

◦ Raised power button or colored button (e.g., red for power) afford immediate attention & action.

◦ AC Remote :-

◦ Buttons with carved icons (snowflake for cool, sun for heat) afford selection for the respective function.

◦ Rocker switches/toggles afford sliding or switching b/w modes.

Good affordance implies users can easily

guess what interaction is possible: pressing for action, toggling for switching, etc.

→ Signifiers = Communicate what actions are possible and where to interact.

- TV Remote:
 - Clear labels ("Menu", "Power", "Mute"), universally recognized symbols (red dot for record, speaker icon for mute).
 - LEDs or backlighting signal that a button has registered an input.
- AC Remote:
 - LCD screen icons & changing display numbers signify changes in temperature & mode.
 - Flashing symbols after a button press act as signifiers, indicating selected options are active.

Effective signifiers make functions visible and actionable, reducing guesswork for users.

2. Discuss how conflicts between UI/UX principles (e.g., simplicity vs. feature richness or consistency vs. innovation) can affect interface design, and suggest strategies to balance them.

Common Conflicts

- Simplicity vs. Feature Richness: Adding features can clutter and overwhelm, but too much simplicity may hide important functions.
- Consistency vs. Innovation: Consistency aids learnability; innovation can confuse users if changes are unexpected.

Impact on Design

- Cluttered interfaces become hard to

navigate; oversimplified ones may frustrate users needing advanced features.

- Innovations without clear signifiers can confuse users and increase errors

Strategies to balance

- Progressive Disclosure :- Show adv. features only when needed.
- User customization :- Let users tailor access to frequently used functions.
- Clear visual hierarchy :- Highlight common actions while de-emphasizing secondary ones.
- Guided Onboarding :- Introduce new elements gradually with help cues.
- Consistent Core :- Keep basic controls uniform while innovating less critical features.
- User feedback :- Iterate based on real user experiences.